

|    | Q1&2 | Q3 | Q4&5 | Q6<br>a)b)c) | Q6<br>d) | Q7a) | Q7b) | Q8a) | Q8b) | Q8c) |     |
|----|------|----|------|--------------|----------|------|------|------|------|------|-----|
| P4 | /2   |    |      | /10          |          |      |      | /2   |      |      | /14 |
| H5 |      |    |      |              |          |      | /3   |      | /3   | /7   | /13 |
| H6 |      |    |      |              |          | /9   |      |      |      |      | /9  |
| H7 |      | /1 | /2   |              | /2       |      |      |      |      |      | /5  |
|    |      |    |      |              |          |      |      |      |      |      | /41 |

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## Section I

5 marks

Attempt Questions 1–5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1. The parabola  $y = kx^2 - 2x - 1$  is negative definite. What are the possible values of  $k$ ?

(A)  $k < -1$

(C)  $k < 1$

(B)  $k \leq -1$

(D)  $k \leq 1$

2. What is the equation of a parabola that has a vertex of  $(0,0)$  and a focus  $(2,0)$ ?

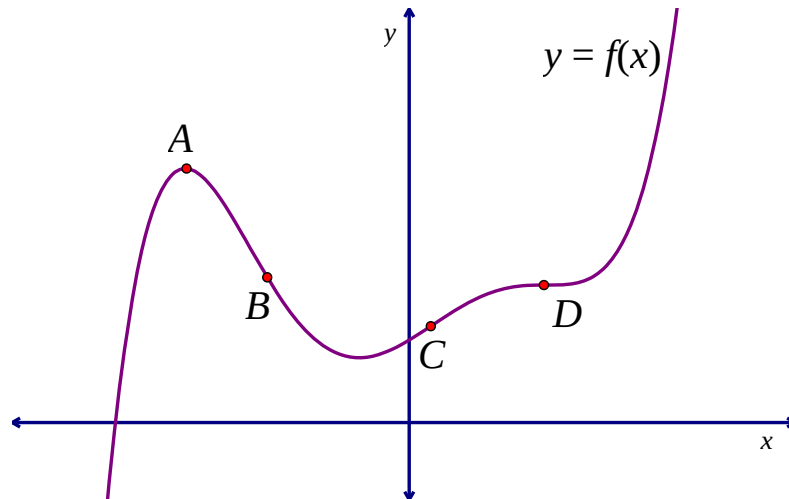
(A)  $(x-2)^2 = -8y$

(C)  $(y-2)^2 = 8x$

(B)  $x^2 = -8y$

(D)  $y^2 = 8x$

3. Below is the graph of  $y = f(x)$



Which point has  $f'(x) < 0$  and  $f''(x) = 0$

(A) A

(C) C

(B) B

(D) D

4. What is the primitive function of  $f(x) = 2x^2 + \frac{5}{x^2}$ ?

(A)  $4x - \frac{10}{x^3} + C$

(C)  $\frac{2x^3}{3} + \frac{5}{3x^3} + C$

(B)  $6x^3 - \frac{5}{x} + C$

(D)  $\frac{2x^3}{3} - \frac{5}{x} + C$

5. If  $x^2 + 2x + 2 \equiv A(x^2 - 1) + B(x + 1) + 3C$ , what is the value of  $C$ ?

(A)  $C = -\frac{1}{3}$

(C)  $C = \frac{5}{3}$

(B)  $C = -\frac{5}{3}$

(D)  $C = \frac{1}{3}$

**End of Section I**

## Section II

36 marks

Attempt Questions 6–8

Allow about 42 minutes for this section

Answer each question in the appropriate writing booklet. Extra pages are available.

In Questions 6 - 8, your responses should include relevant mathematical reasoning and/or calculations.

**Question 6** (12 Marks) Use the Question 6 Writing Booklet.

- (a) The equation  $3x^2 - 4x - 2 = 0$  has roots  $\alpha$  and  $\beta$ .  
Find the value of:

(i)  $\alpha + \beta$  1

(ii)  $\alpha\beta$  1

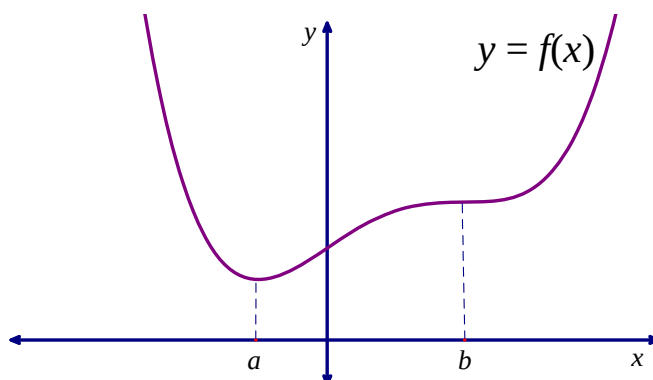
(iii)  $\alpha^2 + \beta^2$  2

- (b) Use the substitution  $u = x^2 - 2$  or otherwise to solve 3

$$4(x^2 - 2)^2 - 29(x^2 - 2) + 7 = 0$$

- (c) The point  $P(x, y)$  moves so that it is equidistant from  $(2, 1)$  and  $(3, 3)$ . 3  
Find the equation of the locus of  $P$ .

- (d) Below is the graph of  $y = f(x)$  2



Sketch the graph of  $y = f'(x)$ , clearly indicating the points where  $x = a$  and  $x = b$ .

**End of Question 6**

**Question 7** (12 Marks) Use the Question 7 Writing Booklet.

- (a) Consider the function  $f(x) = x^3 - 3x^2$ .
- (i) Find the coordinates of any stationary points of  $f(x)$  and determine their nature. **3**
  - (ii) Find the coordinates of any inflexion points. **2**
  - (iii) Hence sketch the graph of  $y = f(x)$ . Clearly indicate the stationary points, any points of inflexion and any intercepts with the axes. **3**
  - (iv) Hence determine the values of  $k$  so that the equation  $x^3 - 3x^2 = k$  has three solutions. **1**
- (b)
- (i) Express the equation of the parabola  $x^2 + 2x + 8y + 25 = 0$  in the form: **2**  

$$(x - h)^2 = -4a(y - k)$$
  - (ii) Hence find the equation of the directrix of the parabola. **1**

**End of Question 7**

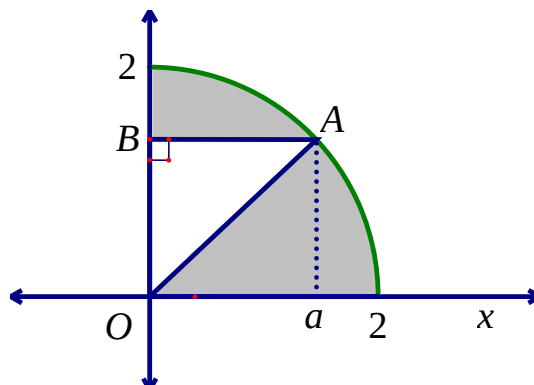
**Question 8\_** (12 Marks) Use the Question 8 Writing Booklet.

- (a) Find the values of  $k$  such that  $y = x^2 + 8x - 5$  and  $y = -2x^2 + x + k$  Do not intersect. 2

- (b) The derivative of a function  $f(x)$  is  $f'(x) = 2x + 1$ . The line  $y = -3x + 2$  is a tangent to the graph of  $f(x)$ . 3

Find the function  $f(x)$ .

- (c) The diagram represents the curve  $x^2 + y^2 = 4$  where  $x \geq 0$  and  $y \geq 0$ . The point  $A$  lies on the curve and  $B$  is the foot of the perpendicular from  $A$  to the  $y$ -axis. The  $x$ -coordinate of  $A$  is  $a$ .



- (i) Show that the shaded area  $S$  in terms of  $a$  is 2

$$S = \pi - \frac{a\sqrt{4-a^2}}{2}$$

- (ii) Show that  $\frac{dS}{da} = \left( \frac{a^2 - 2}{\sqrt{4-a^2}} \right)$  2

- (iii) Find the minimum area of the shaded region. 3

**End of Paper**

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## SECTION I

Q1 A

$$y = kx^2 - 2x - 1$$

$$\Delta = (-2)^2 - 4(k)(-1)$$

Negative definite when:

$$\Delta < 0 \quad \text{leading coefficient} < 0$$

$$4 + 4k < 0$$

$$4k < -4$$

$$k < -1$$

Q2 D

$$\text{Form is } y^2 = 4ax$$

Focal length is 2

$$\text{Solution is } y^2 = 8x$$

Q3 B

Gradient is negative at B and it is a change in concavity.

Q4 D

$$f(x) = 2x^2 + \frac{5}{x^2}$$

$$f(x) = 2x^2 + 5x^{-2}$$

$$F(x) = 2x^3 - 5x^{-1} + C$$

$$F(x) = 2x^3 - \frac{5}{x} + C$$

Q5 D

$$x^2 + 2x + 2 \equiv A(x^2 - 1) + B(x + 1) + 3C$$

Sub  $x = -1$  into both LHS and RHS

$$(-1)^2 + 2(-1) + 2 = A((-1)^2 - 1) + B((-1) + 1) + 3C$$

$$1 - 2 + 2 = 3C$$

$$C = \frac{1}{3}$$

## SECTION II

### Question 6

$$3x^2 - 4x - 2 = 0$$

$$\alpha + \beta = -\frac{b}{a}$$

a)i)

$$= -\frac{-4}{3} = \frac{4}{3}$$

$$\alpha\beta = \frac{c}{a}$$

a)ii)

$$= \frac{-2}{3}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$
$$= \left(\frac{4}{3}\right)^2 - 2\left(-\frac{2}{3}\right)$$

a)iii)

$$= \left(\frac{16}{9}\right) + \left(\frac{4}{3}\right)$$
$$= \frac{28}{9}$$

b)

$$4(x^2 - 2)^2 - 29(x^2 - 2) + 7 = 0$$

$$u = x^2 - 2$$

$$4(u)^2 - 29(u) + 7 = 0$$

$$(4u - 1)(u - 7) = 0$$

$$u = \frac{1}{4} \quad u = 7$$

$$x^2 - 2 = \frac{1}{4} \quad x^2 - 2 = 7$$

$$x^2 = \frac{9}{4} \quad x^2 = 9$$

$$x = \pm \frac{3}{2} \quad x = \pm 3$$

c)

let the distance from  $P(x, y)$  to  $(2, 1)$  be  $d_1$

let the distance from  $P(x, y)$  to  $(3, 3)$  be  $d_2$

$$d_1 = \sqrt{(x-2)^2 + (y-1)^2}$$

$$d_2 = \sqrt{(x-3)^2 + (y-3)^2}$$

$$\text{If } d_1 = d_2$$

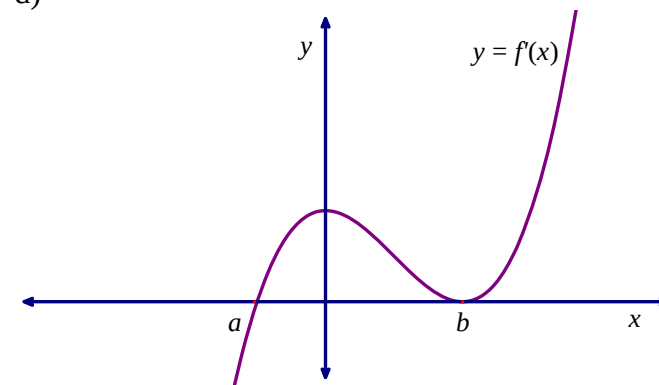
$$(x-2)^2 + (y-1)^2 = (x-3)^2 + (y-3)^2$$

$$x^2 - 4x + 4 + y^2 - 2y + 1 = x^2 - 6x + 9 + y^2 - 6y + 9$$

Locus of  $P$

$$2x + 4y - 13 = 0$$

d)



### Question 7

a) i)

$$f(x) = x^3 - 3x^2$$

$$f'(x) = 3x^2 - 6x$$

$$f''(x) = 6x - 6$$

Stationary points where

$$f'(x) = 0$$

$$f'(x) = 3x^2 - 6x$$

$$0 = 3x^2 - 6x$$

$$0 = 3x(x - 2)$$

Stationary Points at

$$x = 0 \quad \text{and} \quad x = 2$$

*y* - Values

$$f(0) = (0)^3 - 3(0)^2 = 0$$

$$f(2) = (2)^3 - 3(2)^2 = -4$$

Nature of stationary points

Test in 2<sup>nd</sup> derivative

$$f''(0) = 6(0) - 6 = -6$$

∴ MAX turning point at (0,0)

$$f''(2) = 6(2) - 6 = 6$$

∴ MIN turning point at (2,-4)

a) ii)

Inflection point at change in concavity

Possible point of inflexion where

$$f''(x) = 0$$

$$0 = 6x - 6$$

$$6 = 6x$$

$$x = 1$$

Test either side of  $x = 1$  in second derivative

|          |    |   |   |
|----------|----|---|---|
| $x$      | 0  | 1 | 2 |
| $f''(x)$ | -6 | 0 | 6 |

Change of concavity at  $x = 1$

$$\text{When } x = 1 \quad f(1) = (1)^3 - 3(1)^2 = -2$$

Point of inflexion at (1,-2)

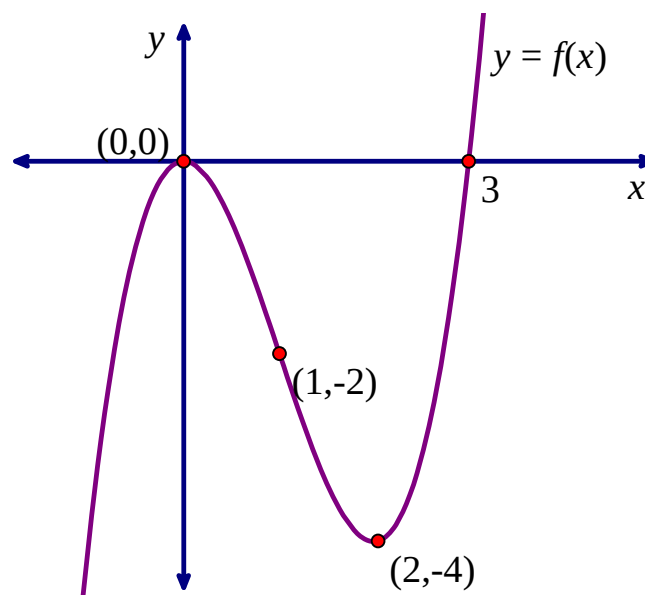
a) iii)

$x$  - intercepts

$$0 = x^3 - 3x^2$$

$$0 = x^2(x - 3)$$

$$x = 0 \quad \text{and} \quad x = 3$$



a) iv)

$$x^3 - 3x^2 = k$$

From the graph the equation will have three solutions if  $-4 < k < 0$

b) i)

$$x^2 + 2x + 8y + 25 = 0$$

Complete the square.

$$x^2 + 2x + 1 = -8y - 24$$

$$(x + 1)^2 = -8(y + 3)$$

b) ii)

Vertex is at (-1,-3)

Focal Length is 2

Concave Down Parabola

Directrix is at

$$y = -1$$

### Question 8

a)

Solve the two  $y = x^2 + 8x - 5$  and  $y = -2x^2 + x + k$  equations simultaneously

$$-2x^2 + x + k = x^2 + 8x - 5$$

$$0 = 3x^2 + 7x - k - 5$$

No intersections if  $\Delta < 0$

$$\Delta = 7^2 - 4 \times 3 \times (-k - 5)$$

$$\Delta = 49 + 12k + 60$$

$$\Delta = 109 + 12k$$

$$109 + 12k < 0$$

$$12k < -109$$

$$k < \frac{-109}{12}$$

$$k < -9\frac{1}{12}$$

b)

$$f'(x) = 2x + 1$$

$$F(x) = x^2 + x + C$$

METHOD 1

As  $y = -3x + 2$  is a tangent then

where  $f'(x) = -3$  will be the x-value of the point of intersection.

Let

$$-3 = 2x + 1$$

$$-4 = 2x$$

$$x = -2$$

y- Value of the point of intersection is

$$y = -3(-2) + 2$$

$$y = 8$$

$\therefore (-2, 8)$  lies on the curve

$$y = x^2 + x + C$$

$$\therefore 8 = (-2)^2 + (-2) + C$$

$$8 = 4 - 2 + C$$

$$C = 6$$

$$\therefore f(x) = x^2 + x + 6$$

METHOD 2

The tangent  $y = -3x + 2$  and function

$F(x) = x^2 + x + C$  will have only one intersection point. Solving simultaneously will only provide one solution.

$$x^2 + x + C = -3x + 2$$

$$x^2 + 4x + C - 2 = 0$$

$$\Delta = 4^2 - 4 \times 1 \times (C - 2)$$

$$\Delta = 16 - 4C + 8$$

$$\Delta = 24 - 4C$$

One solution when  $\Delta = 0$

$$24 - 4C = 0$$

$$C = 6$$

$$\therefore f(x) = x^2 + x + 6$$

c) i)

$$\text{Area}_{\text{Shaded}} = \text{Area}_{\text{QuarterCircle}} - \text{Area}_{\text{Triangle}}$$

$$\text{Area}_{\text{Shaded}} = \frac{\pi \times r^2}{4} - \frac{1}{2} \times OB \times BA$$

$$OB = \sqrt{4 - BA^2} \quad \text{by Pythagoras' Theorem}$$

$$\text{Area}_{\text{Shaded}} = \frac{\pi \times 2^2}{4} - \frac{1}{2} \times \sqrt{4 - a^2} \times a$$

$$\text{Area}_{\text{Shaded}} = \pi - \frac{a\sqrt{4 - a^2}}{2}$$

c) ii)

$$S = \pi - \frac{a\sqrt{4 - a^2}}{2}$$

$$\frac{dS}{da} = u'v + v'u$$

$$= -\frac{1}{2} \left[ 1 \times \sqrt{4 - a^2} + \frac{1}{2} (4 - a^2)^{-\frac{1}{2}} (-2a)(a) \right]$$

$$= -\frac{1}{2} \left[ \sqrt{4 - a^2} - \frac{a^2}{\sqrt{4 - a^2}} \right]$$

$$= -\frac{1}{2} \left[ \frac{4 - a^2 - a^2}{\sqrt{4 - a^2}} \right]$$

$$= \frac{a^2 - 2}{\sqrt{4 - a^2}}$$

C iii)

Possible min when  $\frac{dS}{dt} = 0$

$$0 = \frac{a^2 - 2}{\sqrt{4 - a^2}}$$

$$0 = a^2 - 2$$

$$a^2 = 2$$

$$a = \pm\sqrt{2}$$

But  $a > 0$

When  $a = \sqrt{2}$  the shaded area is a possible minimum.

Test either side of  $a = \sqrt{2}$  in  $\frac{dS}{dt}$

| $a$             | 1.4                                          | $\sqrt{2}$ | 1.5                                          |
|-----------------|----------------------------------------------|------------|----------------------------------------------|
| $\frac{dS}{dt}$ | $\frac{(1.4)^2 - 2}{\sqrt{4 - (1.4)^2}} < 0$ | 0          | $\frac{(1.5)^2 - 2}{\sqrt{4 - (1.5)^2}} > 0$ |

$\therefore$  When  $a = \sqrt{2}$  the shaded area is a minimum

When  $a = \sqrt{2}$

$$S = \pi - \frac{\sqrt{2}\sqrt{4 - (\sqrt{2})^2}}{2}$$

$$= \pi - \frac{\sqrt{2}\sqrt{2}}{2}$$

$$= \pi - 1 \text{ units}^2$$